

AMENDMENT NO. 1 NOVEMBER 2024
TO
IS 3224 : 2021 VALVE FOR COMPRESSED GAS CYLINDERS EXCLUDING
LIQUEFIED PETROLEUM GAS (LPG) CYLINDERS — SPECIFICATION
(Fourth Revision)

(Page 1, clause 1.2, line 1) — Substitute the following for the existing:

‘This standard gives the details of the dimensions of inlet taper threads as well as parallel threads and outlet of the valves so as to ensure interchangeability.’

(Page 3, clause 3.61, title) — Substitute ‘**Valve Inlet/Valve Stem Connection**’ for ‘Valve Inlet Connection’.

(Page 5, Fig. 2, title) — Substitute the following for the existing:

‘ILLUSTRATION OF PRESSURE SEAL VALVE, NON-METALLIC SEAL (WHEEL OPERATED)’.

(Page 7, clause 5.2) — Delete.

(Page 8, clause 5.3.1) — Substitute the following for the existing:

‘The tensile strength and elongation of the material of the valve body determined according to IS 1608 (Part 1) shall be respectively at least 392 MPa (40 kgf/mm²) and minimum 18 percent measured on a gauge length $5.65\sqrt{S_0}$ (S_0 being the original area of cross-section) except for valves used for CO₂ and N₂O for which the minimum tensile strength of the material of the valve body according to IS 1608 (Part 1) shall be 343 MPa (35 kgf/mm²) and minimum elongation shall be 18 percent on a gauge length $5.65\sqrt{S_0}$.’

(Page 8, clause 5.3.2, title) — Substitute ‘*Izod Impact Test*’ for ‘*Impact Strength*’.

[Page 9, Table 1, Sl No. xxii), col (2)] — Substitute ‘Difluoromethane’ for ‘Difloro methane’.

[Page 10, Table 1, Sl No. li)] — Insert the following at the end:

Sl No.	Gas		Designation of Screw Thread Outlet (Parallel Threads) ¹⁾	Pitch	Outlet No.	Width Across Flat of the Square of Spindle ²⁾
	Name	Chemical Symbol				
(1)	(2)	(3)	(4)	(5)	(6)	(7)
lii)	Bio-CNG	Bio-CNG ⁵⁾	G5/8-14 TPI-LH	1.814	21	9.5 mm
liii)	Hydrogen (for working pressure up to 300 bar)	H ₂	G5/8-14 TPI-LH	1.814	2	7.1
liv)	Hydrogen (for working pressure > 300 bar and up to 800 bar)	H ₂	1.250-14NGO-LH-INT	1.814	22	-

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(Page 10, Table 1, key) — Insert the following at the end:

‘5 Bio-CNG valves shall be strictly used for bio-gas conforming to IS 16087.’

(Page 10, clause 6.3.2, line 4) — Substitute ‘values’ for ‘valves’.

(Page 10, clause 6.3.2, para 2, line 2) — Substitute ‘shall’ for ‘will’.

[Page 11, Table 2, Sl No. iii), col (4)(g)] — Substitute ‘other’ for ‘pther’.

(Page 12, clause 6.3.3, Note 2) — Substitute the following for the existing:

‘For special designs, the number of cycles shall be defined by the manufacturer on the basis of a specification from the customer or industry based on the likely service conditions. The number of cycles shall be documented in the drawing. The number of cycles shall be 500 for squeeze grip valves.’

[Page 13, Table 3, Sl No. x), col (7)] — Substitute ‘12 to 14’ for ‘15 to 1.7’.

(Page 14, clause 7.2.1) — Substitute the following for the existing:

‘Where the pressure relief device is a bursting disc fitted to the valve of seamless or welded cylinder the bursting pressure of the disc (when tested at temperature of 65 °C) shall not exceed the 110 percent of test pressure of the cylinder for which device is intended and shall be more than the developed pressure of the gas at 65 °C. Bursting disc may be rated at room temperature provided correlation between room temperature and elevated temperature is determined.’

(Page 14, clause 8.1.1, Note, line 2) — Substitute ‘CGA Standard CGA-V-1’ for ‘CGA Standard V-1-2013’.

(Page 16, Fig. 9b, title) — Substitute ‘VALVE INLET’ for ‘VALVE STEM’.

(Page 16, Fig. 9b, Table, first heading) — Substitute ‘Nominal Diameter of Valve’ for ‘Normal Diameter of Valve’.

(Page 16, Fig. 9b, Table) — Substitute the following for the existing:

Sl No.	Normal Diameter of Valve ¹⁾	Valve Inlet Thread			Thread in Cylinder Neck		Length of Valve Inlet Thread	
		d + 0.12	d + 0.12	L_1	d_2 - 0.12	L_2 Min	L_3 Remaining Outside the Cylinder	L_4 Engaged Inside the Cylinder
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
i)	19.8	19.8	17.4	21	19.2	17	5.0	16.0
ii)	28.8	28.8	25.8	26	27.8	22	8.33	17.67

(Page 19, clause 8.1.2, Note) — Substitute the following for the existing:

‘This type of thread also conforms to DIN 477 and ISO 11363-1. The 17 E and 25 E threads have profile normal to the cone.’

(Page 19, clause 8.3.2, line 3) — Substitute ‘manufacturer’ for ‘manufacture’.

(Page 24, clause 9.1.1, line 3) — Delete ‘and outlet connection listed in this clause’.

(Page 26, Figure of Outlet No. 2, title) — Substitute the following for the existing:

‘OUTLET NO. 2 OUTLET CONNECTION FOR ACETYLENE, BUTADIENE, CARBON MONOXIDE, COAL GAS, ETHYLENE, METHANE, DIMETHYL ETHER, ISOBUTYLENE AND HYDROGEN (USED FOR WORKING PRESSURE UP TO 300 BAR)’.

(Page 27, Figure of Outlet No. 5) — Substitute ‘CHLOROTRIFLUOROMETHANE’ for ‘CHLOROTRIFLUOROMETHENE’.

(Page 27, Figure of Outlet No. 5, last line) — Substitute ‘BORON TRIFLUORIDE AND DIFLUOROMETHANE’ for ‘AND BORON TRIFLUORIDE’.

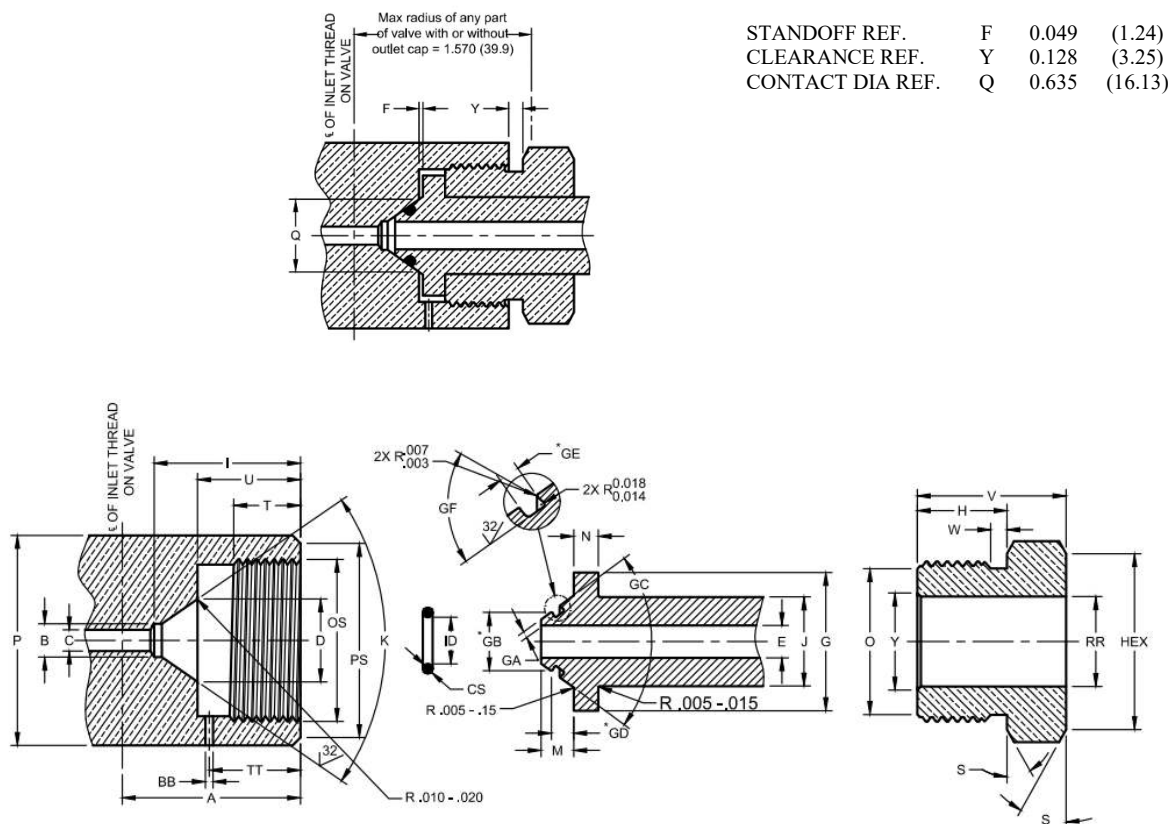
(Page 27, Figure of Outlet No. 6) — Substitute ‘METHYL CHLORIDE’ for ‘METHYLE CHLORIDE’.

(Page 30, Figure of Outlet No. 17) — Substitute ‘CYANOGEN’ for ‘CYNNOGEN’.

(Page 31, Figure of Outlet No. 21, title) — Substitute the following for the existing:

‘OUTLET NO. 21 OUTLET CONNECTION FOR COMPRESSED NATURAL GAS AND BIO-COMPRESSED NATURAL GAS’.

[Page 32, Figure of Outlet No. 21(A)] — Insert the following at the end:



All dimensions in millimetres.

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VALVE OUTLET

<i>Sl No.</i>	<i>Dimension</i>	<i>Symbol</i>	<i>Dimension</i>	
			inch	mm
(1)	(2)	(3)	(4)	(5)
i)	Thread	1.250-14NGO-LH-INT		
ii)	Minor diameter	-	1.172 7 to 1.180 4	29.787 to 89.983
iii)	Pitch diameter	-	1.203 6 to 1.207 6	30.572 to 30.673
iv)	Major diameter	-	1.250 0 <i>Min</i>	31.750 <i>Min</i>
v)	Length	A	1.312 <i>Max</i>	33.33 <i>Max</i>
vi)	Bore depth	B	0.250 <i>Max</i>	6.35 <i>Max</i>
vii)	Drill diameter	C	0.156 <i>Min</i>	3.96 <i>Min</i>
viii)	Bore diameter	D	0.635	16.13
ix)	C'bore depth	I	1.125 <i>Min</i>	28.58 <i>Min</i>
x)	Angle	K	69°	
xi)	Boss diameter	P	1.625 <i>Min</i>	41.28 <i>Min</i>
xii)	Full thread	T	0.600 <i>Min</i>	15.24 <i>Min</i>
xiii)	Bore depth	U	0.793	20.14
xiv)	Bl hole diameter	BB	0.062	1.57
xv)	Bl hole dist.	TT	0.685 to 0.715	17.40 to 18.16
xvi)	Thd chamfer	OS	45° × 1.265	32.13
xvii)	Chamfer diameter	PS	45° × 1.495	37.97

NIPPLE¹⁾

<i>Sl No.</i>	<i>Dimension</i>	<i>Symbol</i>	<i>Dimension</i>	
			inch	mm
(1)	(2)	(3)	(4)	(5)
i)	Drill diameter	E	187 ± 0.60	4.75 ± 1.2
ii)	Shoulder diameter	G	1.055	26.80
iii)	Shank diameter	J	0.675	17.145
iv)	Nose length	M	0.260	6.60
v)	Shoulder length	N	0.177 to 0.192	4.50 to 4.88
vi)	Shank length	Z	1.125 <i>Min</i>	28.58 <i>Min</i>
vii)	Groove depth	GA	0.051 to 0.053	1.30 to 1.34
viii)	Groove diameter ²⁾	GB	0.446 to 0.454	11.33 to 11.53
ix)	Angle	GC	70°	
x)	Groove location ²⁾	GD	0.180	4.57
xi)	Groove width ²⁾	GE	0.075 to 0.079	1.91 to 2.01
xii)	Dovetail angle	GF	64° to 68°	

HEXAGON NUT

<i>Sl No.</i>	<i>Dimension</i>	<i>Symbol</i>	<i>Dimension</i>	
			inch	mm
(1)	(2)	(3)	(4)	(5)
i)	Thread	1.245-14NGO-LH-EXT		
ii)	Major diameter	-	1.245 0 to 1.240 0	31.623 to 31.496
iii)	Pitch diameter	-	1.198 6 to 1.194 6	30.44 to 30.35
iv)	Minor diameter	-	1.157 4 <i>Max</i>	29.39 <i>Max</i>

¹⁾ Nipple may be made from 1.000 (25.40) hex material.

²⁾ Dimension to theoretical intersection.

Sl No.	Dimension	Symbol	Dimension	
			inch	mm
v)	Hex		1.375	34.92
vi)	Shank length	H	0.687 ± 0.15	17.450 ± 38
vii)	Chamfer diameter	O	$30^\circ \times 1.144$	29.06
viii)	Hole diameter	R	0.685 to 0.690	17.40 to 17.52
ix)	Hex chamfer	S	$30^\circ \times 1.375$	34.92
x)	Length	V	1.125 <i>Min</i>	28.57 <i>Min</i>
xi)	Neck	W	0.16×1.144 <i>Min</i>	4.1×29.06 <i>Min</i>
xii)	R chamfer diameter	Y	$45^\circ \times 0.720$ to 0.750	18.29 to 19.05

O-RING

Sl No.	Dimension	Symbol	Dimension	
			inch	mm
(1)	(2)	(3)	(4)	(5)
i)	Size (Ref)	2 - 012		
ii)	Cross section	CS	0.067 to 0.073	1.71 to 1.85
iii)	Inside diameter	ID	0.364	(9.25)
iv)	Durometer, shore A	-	85 to 95	-

OUTLET NO. 22 OUTLET CONNECTION FOR HYDROGEN GAS SERVICE USED FOR
WORKING PRESSURE GREATER THAN 300 BAR AND UPTO 800 BAR

[Page 33, clause **10.1.5(a)**, line 1] — Substitute the following for the existing:

‘Repeat of valve impact test, valve burst pressure test, endurance test and subsequent leak tightness test;’

[Page 33, clause **10.1.5(n)**, title] — Delete ‘/Integration’.

[Page 33, clause **10.1.5(q)**, line 1 and line 3] — Substitute ‘shall’ for ‘will’.

[Page 33, clause **10.2(g)**] — Substitute ‘Outlet sizes’ for ‘Outlet sizes and gas services’.

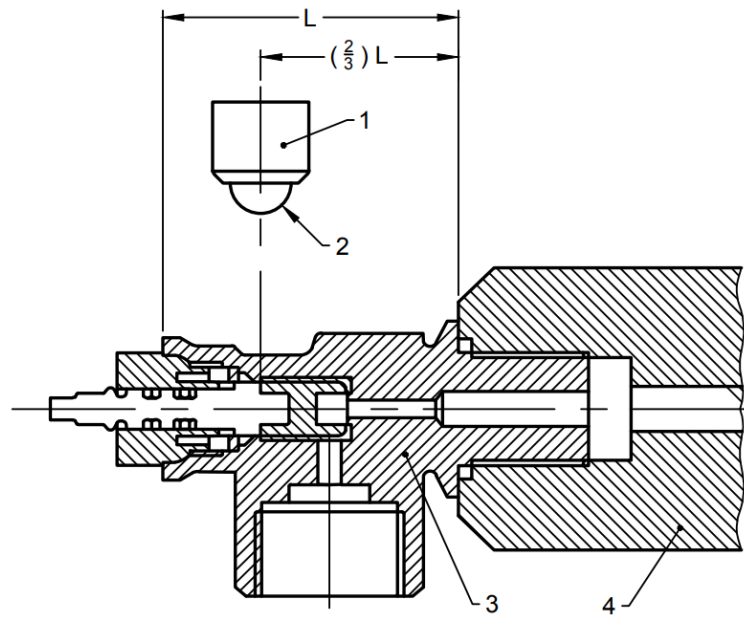
(Page 34, clause **10.3.2**) — Substitute ‘ $P_{vt} = P_w$ ’ for ‘ $P_{vt} \geq P_w$ ’.

(Page 34, clause **10.4.1**, line 2) — Substitute the following for the existing:

‘The burst pressure test shall be carried out with the valve seat in open position with valve outlet/filling connection(s) plugged.’

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(Page 35, Fig. 13) — Substitute the following for the existing:



Key

- 1 Plummet weight
- 2 Hardened steel ball, diameter 13 mm
- 3 Test sample
- 4 Test fixture or cylinder
- a Longitudinal axis

(Page 35, Table 13, title) — Substitute the following for the existing:

‘Table 13 — Recommended Valving Torques for Taper Threaded Valve Stems for Seamless Steel Cylinder and Composite Cylinders with Steel Boss’.

[Page 35, Table 13, col (5) and col (6), title] — Delete sub-heading ‘Seamless Steel Cylinder’.

(Page 36, Table 14) — Substitute the following for the existing:

Table 14 Recommended Valving Torque for Parallel Threaded Valve Stem for Seamless Steel Cylinders and Composite Cylinders with Steel Boss

(Clauses 10.4.2.3, 10.4.3.4 and 10.4.3.5)

Sl No.	Valve Material	Category	Inlet Thread Code	Valving Torque, Nm	
				Min	Max
(1)	(2)	(3)	(4)	(5)	(6)
i)	Copper base alloy, Carbon steel and Stainless steel	a) I	a) 18P	100	130
			b) U12		
		b) II	c) 25P	100	130
			d) SP12		
			e) U18		
			f) 30P		

(Page 36, Table 15) — Substitute the following for the existing:

**Table 15 Recommended Valving Torques of Taper Threaded Valve Stems for Aluminium Alloy
Cylinders and Composite Cylinders with Aluminium Alloy Boss**
(Clauses 10.4.2.3, 10.4.3.4 and 10.4.3.5)

SI No.	Valve Material	Category	Inlet Thread Code	Valving Torque		
				Min Nm	Max Nm	
					Without Cylinder Neck Reinforcement	With Cylinder Neck Reinforcement
(1)	(2)	(3)	(4)	(5)	(6)	(7)
i)	Copper base alloy and carbon steel	a) I	a) 18T	a) 75	a) 95	a) 140
			b) 08N			
			c) 17E			
		b) II	d) 25.4	b) 95	b) 110	b) 180
			e) 12N			
			f) 25E			
NOTE — The torque figures given above are for use with PTFE thread sealant. If different sealant or pressure ranges are introduced, the torque figures given in the table may have to be changed to ensure a gas tight joint.						

(Page 37, clause 10.4.4.3.3, line 2) — Substitute the following for the existing:

‘This value of T_0 shall be applied on the other two test samples then used for determination of T_f , in closing and opening direction.’

(Page 38, clause 10.5.1, line 1) — Substitute ‘shall’ for ‘will’.

[Page 38, clause 10.5.3(f)] — Delete.

(Page 39, clause 10.6.1.11, line 2) — Substitute ‘shall’ for ‘will’.

(Page 40, Fig. 14, Key 1) — Substitute ‘Torque’ for ‘TORQE’.

(Page 40, Fig. 14, Key 8) — Substitute ‘Display’ for ‘DIPLAY’.

(Page 40, Fig. 15) — Substitute ‘Atmospheric pressure’ for ‘ATMOSPHERIC RESSURE’.

[Page 41, clause 10.9(b), line 2] — Substitute ‘ W_c ’ for ‘W’.

(Page 43, clause 11.1.5) — Delete the clause and its sub clause.

(Page 44, clause 11.1.8.4.1) — Delete ‘Test Procedure:’ after heading.

(Page 44, clause 13.3, line 1) — Substitute the following for the existing:

‘All valves shall be subjected to both internal leak tightness and external leak tightness at room temperature at minimum working pressure (P_w).’

(Page 45, clause 13.3, para 2) — Substitute the following for the existing:

‘For valves equipped with pressure relief device, testing shall be done at minimum 0.8 times the minimum rated burst pressure of pressure relief device.’

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(Page 45, Table 22) — Substitute the following for the existing:

Table 22 Scale of Sampling

(Clause 13.4.1)

Sl No. (1)	Batch Size (2)	Sample Size (3)
i)	3 up to including 3 000	2

(Page 45, clause 13.4.5, line 2) — Substitute ‘shall’ for ‘will’.

(Page 45, clause 13.4.6, line 2) — Substitute ‘shall’ for ‘will’.

(Page 45, clause 13.4.5) — Substitute ‘**13.5 Checking of Inlet Connection**’ for ‘13.4.5 Checking of Inlet Connection’.

(Page 45, clause 13.4.6) — Substitute ‘**13.6 Checking of Outlet Connection**’ for ‘13.4.6 Checking of Outlet Connection’.

(Page 45, clause 13.4.7) — Substitute ‘**13.7 Checking of Other Dimensions**’ for ‘13.4.7 Checking of Other Dimensions’.

(Page 45, clause 13.4.8) — Substitute ‘**13.8 Scale of Sampling**’ for ‘13.4.8 Scale of Sampling’.

(Page 45, clause 13.4.8.1) — Substitute ‘**13.8.1**’ for ‘13.4.8.1’.

(Page 45, clause 13.4.8.2) — Substitute ‘**13.8.2**’ for ‘13.4.8.2’.

(Page 45, clause 13.4.8.3) — Substitute ‘**13.8.3**’ for ‘13.4.8.3’.

[Page 45, clause 14.1(e)] — Substitute the following for the existing:

‘Working pressure, MPa/bar/(kgf/cm²)’.

[Page 46, clause 14.1(m), line 2] — Delete ‘or PRD’.

(Page 47, Annex A, left side column, point 4) — Substitute ‘IS 1608 (Part 1) : 2022/ISO 6892-1 : 2019 Metallic materials — Tensile testing: Part 1 Method of test at room temperature (*fifth revision*)’ for ‘1608 (Part 1) : 2018 Metallic materials — Tensile testing: Part 1 Method of test at room temperature (*fourth revision*)’.

(Page 47, Annex A, right side column, point 7) — Substitute ‘IS/ISO 11114-1 : 2020 Gas cylinders — Compatibility of cylinder and valve materials with gas contents: Part 1 Metallic materials (*first revision*)’ for ‘IS/ISO 11114-1 : 2012 Transportable gas cylinders — Compatibility of cylinder and valve materials with gas contents — Metallic materials’.

(Page 47, Annex A) — Insert the following at the end:

<i>IS No./Other Standards</i>	<i>Title</i>
IS 2102 (Part 1) : 1993/ ISO 2768-1 : 1989	General tolerances: Part 1 Tolerances for linear and angular dimensions without individual tolerance indications (<i>third revision</i>)
IS 3745 : 2006	Yoke type valve connections for small medical gas cylinders — Specification (<i>second revision</i>)

<i>IS No./Other Standards</i>	<i>Title</i>
IS 7302 : 2018	Valve fittings for self contained breathing apparatus (SCBA) and self contained underwater breathing apparatus (SCUBA) — Specification (<i>first revision</i>)
IS 8776 : 1988	Specification for valve fittings for use with liquefied petroleum gas (LPG) cylinders up to and including 5 litre water capacity (<i>first revision</i>)
IS 12300 : 1988	Valve fittings for refrigerant cylinders — Specification
IS 16087 : 2016	Biogas (biomethane) — Specification (<i>first revision</i>)
IS 16988 : 2018	Compressed natural gas cylinder valve integrated with solenoid operation (remotely controlled) for automotive use — Specification
ISO 10297 : 2014	Gas cylinders — Cylinder valves — Specification and type testing
ISO 11363-1 : 2018	Gas cylinders — 17 E and 25 E taper threads for connection of valves to gas cylinders — Part 1: Specifications
ANSI/ASME B1.2 : 1983	Gages and gaging for unified screw threads
ANSI/ASME B1.20-1 : 1983	Pipe threads, general purpose (inch)
BS 341-1 : 2022	Gas cylinders — Cylinder valves: Part 1 Threads for connection of valves to gas cylinders — Specification
DIN 477 : 2021	Gas cylinder valves for cylinder test pressures up to 300 bar — Valve inlet and outlet connections
CGA V-1 : 2023	Standard for compressed gas cylinder valve outlet and inlet connections’.

(MED 16)

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